

Estuti Shukla

CURRICULUM VITÆ • SEPTEMBER 16, 2026

Department of Physics
Institute for Gravitation and the Cosmos
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RESEARCH INTERESTS

Numerical relativity and gravitational-wave modeling for compact binary objects, with a focus on using artificial intelligence to accelerate and improve the accuracy of numerical simulations; black hole perturbation theory and ringdown physics.

EXPERIENCE

Graduate Student Researcher

The Pennsylvania State University — Advisor: Prof. David Radice

Aug. 2022 – Present

University Park, PA

- Developing deep learning methods to improve efficiency and accuracy of numerical relativity simulations
- Performed high-resolution simulations of spinning compact binary coalescence

Visiting Student Researcher

ICTS–TIFR Bangalore — Advisor: Prof. Prayush Kumar

Sept. 2021 – Aug. 2022

Bangalore, India

- Studied properties of dynamical horizons using numerical relativity simulations with the SpEC code

Research Intern

Johns Hopkins University — Advisor: Dr. Vishal Baibhav & Prof. Emanuele Berti

Oct. 2020 – Aug. 2022

Remote

- Investigated nonlinearities associated with ringdown of binary black hole mergers

Summer Research Intern

University of Massachusetts Dartmouth — Advisor: Prof. Gaurav Khanna & Prof. Scott Field

Jun. – Sept. 2020

Remote

- Added remnant Bondi mass and kick velocity functionalities to an open-source EMRI surrogate model
- Credited as a contributor to the public software package

EDUCATION

The Pennsylvania State University

Ph.D. in Physics

State College, USA

Aug. 2022 – Present

- Advisor: Prof. David Radice
- Research: Numerical relativity, AI4Science, black hole perturbation theory

Indian Institute of Science Education and Research Kolkata

B.S.–M.S. in Physics — Major GPA: 9.03/10.00

Kolkata, India

2016 – 2021

- Master's Thesis: *Static Stars in Cosmological Backgrounds*, advisor Prof. Rajesh Nayak
- DST Inspire Scholarship (Govt. of India, 2016 - 2021)

PUBLICATIONS

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- [1] R. Gamba, D. Chiaramello, **E. Shukla**, and S. Albanesi, “Spin the black circle II: tidal heating and torquing of a rotating black hole by a test mass on generic orbits,” submitted to *Phys. Rev. D*, arxiv:603.28982 (2026)
 - [2] **E. Shukla**, A. Rashti, R. Gamba, D. Radice, and K. Chandra, “GR-Athena++ simulations of spinning binary black hole mergers,” *Class. Quant. Grav.* **43**, 065019 (2026).
 - [3] R. Perna, O. Gottlieb, **E. Shukla**, and D. Radice, “Connecting GRBs from binary neutron star mergers to nuclear properties of neutron stars,” *Phys. Rev. D* **111**, 063015 (2025).
 - [4] M.H.-Y. Cheung, V. Baibhav, E. Berti, V. Cardoso, G. Carullo, R. Cotesta, W.D. Pozzo, F. Duque, T. Helfer, **E. Shukla**, and K.W.-K. Wong, “Nonlinear effects in black hole ringdown,” *Phys. Rev. Lett.* **130**, 081401 (2023).

TALKS & CONFERENCES

Invited Talks

Strong Gravity Seminar, Penn State	Sept. 2026
AstroChat Talk, UC Berkeley	July 2026
Primordial Universe and Gravity (PUG) Seminar, IGC, Penn State	Apr. 2024
UMass Dartmouth Physics Colloquium Series (<i>virtual</i>)	Sept. 2020

Contributed Talks

AstroAI Workshop 2026, Harvard University, MA	Jun. 2026
26th Eastern Gravity Meeting, Cornell University, Ithaca, NY	May 2026
AI for Gravitational Waves Workshop, CERN, Geneva (<i>Poster</i>)	May 2026
Neighbourhood Workshop, Penn State, University Park	Apr. 2026
35th Midwest Relativity Meeting & Eric Fest, U. Guelph, Canada	Nov. 2025
ICDS Symposium, Penn State, University Park (<i>Poster</i>)	Sept. 2025
Scientific Machine Learning for GW Astronomy, ICERM, RI (<i>Poster</i>)	Jun. 2025
IGC Alumni Event, Penn State, University Park (<i>Poster</i>)	May 2025
APS Global Physics Summit 2025, Denver, CO	Mar. 2025

Schools & Workshops

IAIFI Summer School, UMass Boston (<i>virtual</i>)	Aug. 2026
Deep Learning for Science School, Berkeley, CA (<i>Poster</i>)	Jul. 2026
AthenaK Summer School, Penn State, University Park	Jul. 2026
Neighbourhood Workshop, Penn State, University Park (<i>attendee</i>)	2023 – 2025
Simulating Extreme Spacetimes with SpEC and SpECTRE, ICERM, RI	Aug. 2024
DTP TALENT Summer School on Nuclear Astrophysics, Trento, Italy	Jul.-Aug. 2024
N3AS Summer School in Multi-messenger Astrophysics, UC Santa Cruz	Jul. 2023
ICTS School on Physics of Early Universe, ICTS-TIFR, Bangalore	Jan. 2022
ICTS Summer School on Gravitational-Wave Astronomy, ICTS-TIFR Bangalore (<i>virtual</i>)	Jul. 2021
Gravitational Wave Summer School on Numerical Relativity, ICTS-TIFR Bangalore (<i>virtual</i>)	May 2020
Finesse Interferometer Modelling Hackathon, IUCAA, Pune (<i>winner</i>)	Mar. 2020
National Science (Vijayoshi) Camp, DST, Gov. of India, IISER Kolkata	Dec. 2016

AWARDS & HONORS

APS DGRAV Student Travel Award, APS Global Summit	2025
EUSTIPEN Travel Award, Trento, Italy	2024
Homer F. Braddock Scholarship, Penn State University	2023
DST Inspire Scholarship, Government of India	2016-2021

TECHNICAL SKILLS

Programming (Advanced): Python, PyTorch, $\text{\LaTeX}$, C/C++, Mathematica
Programming (Intermediate): R, MATLAB
Tools: Git, GNUplot, Origin, Linux

TEACHING AND MENTORSHIP

Teaching Assistant , Penn State University	2022-2024
• PHYS 212 General Physics: Electricity and Magnetism • PHYS 250 Introductory Physics 1	
Teaching Assistant , IISER Kolkata	2020-2021
• General Theory of Relativity & Cosmology (w/ Prof. Rajesh Nayak) • Intermediate Quantum Mechanics (w/ Prof. Soumitro Banerjee)	
Undergraduate Student Mentor	
• Ben Jhonson, Penn State University • Patrick A Bush, San Diego State University (graduate student at OSU)	

PROFESSIONAL ACTIVITIES

Journal Referee

- Classical and Quantum Gravity

Member

- LIGO-Virgo-KAGRA Consortium (2022-)
- American Physical Society Division of Gravitational Physics (2023-)
- Simulating eXtreme Spacetimes Collaboration (2021-2022)

OUTREACH AND SERVICE

Lecture , State College Area High School	Apr 2026
Professional Development Co-Chair , Physics and Astronomy for Women+ (PAW+) at Penn State	2024-2027
Physics Graduate Student Representative , Penn State	2025-2026
Board Member , Physics Graduate Student Association (PGSA) at Penn State	2025-2026
Delegate for Eberly College of Science , Graduate & Professional Student Association, Penn State	2025-2026
Founder of IKQRAAR , official LGBTQ+ support platform at IISER Kolkata	2021-2022
Founder of Trekking & Adventure Club , IISER Kolkata	2020-2021
Founding Member of Student Alumni Cell , IISER Kolkata	2019-2020
Sponsorship Head of Inquivesta , college science festival, IISER Kolkata	2019-2020

REFERENCES

Prof. David Radice

Associate Professor at the Departments of Physics and Astronomy & Astrophysics
Pennsylvania State University
email: dur566@psu.edu

Prof. Romit Maulik

Assistant Professor of Mechanical Engineering
Purdue University
email: rmaulik@purdue.edu

Prof. B.S. Sathyaprakash

Elsbach Professor of Physics and Professor of Astronomy and Astrophysics
Pennsylvania State University
email: bss25@psu.edu

Dr. Rossella Gamba

Postdoc
University of California Berkeley
email: rgamba@berkeley.edu